

Introduction to Biophysics

Technische Universität München

Wintersemester 2025/2026

Prof. Julia Herzen, Prof. Andreas Bausch, Prof. Ulrich Gerland

Solutions to Exercise Sheet 7

Problem 7.1

A protein's primary structure is defined solely by its amino acid sequence, and is constructed by peptide bonds between adjacent amino acid residues. Secondary structure results from hydrogen bonding along the polypeptide backbone, resulting in alpha-helices and beta-pleated sheets. Tertiary structure results from hydrogen bonding between R groups, hydrophobic interactions, and disulfide bridges; these interactions create the three-dimensional structure of the molecule. Finally, quaternary structure arises from the joining of multiple subunits to create a functional protein complex.

Problem 7.2

The nonpolar amino acids are buried in the center of the protein due to their hydrophobic nature.

Not all proteins have a quaternary structure; quaternary structure refers to the arrangement and number of subunits, and not every protein has multiple subunits. Disulfide bonds make proteins less susceptible to unfolding; typically, they will link beta-sheets, alpha-helices, and loops, which means that they primarily maintain tertiary structure, not secondary, which refers to local conformations, and is maintained largely by hydrogen bonds. Protein charges do change with pH; as a solution's pH increases, acidic groups on proteins deprotonate. A protein's function depends, however, very much on its spatial conformation. Its native conformation permits it to recognize and bind to specific molecules, and thus perform its specific function.

Problem 7.3

we first determine the free-energy change represented by the binding to the first protein.

$$\Delta G^\circ = -RT \ln K$$

$$\Delta G^\circ = -(1.98 \times 10^{-3} \text{ kcal/K mole}) (310 \text{ K}) \ln(5 \times 10^9)$$

$$\Delta G^\circ = -13.7 \text{ kcal/mole}$$

The free-energy change associated with binding to the second protein is $-13.7 \text{ kcal/mole} + 2.8 \text{ kcal/mole} = -10.9 \text{ kcal/mole}$. Thus, the equilibrium constant for binding to the second protein is:

$$\begin{aligned} -10.9 \text{ kcal/mole} &= -(1.98 \times 10^{-3} \text{ kcal/K mole}) (310 \text{ K}) \ln k \\ k &= 5 \times 10^7 \end{aligned}$$

Problem 7.4

We first look at the diffusion equation $\frac{\partial c(\vec{r}, t)}{\partial t} = -D \vec{\nabla}^2 c(\vec{r}, t)$. As we are looking at a steady state, the time derivative of the concentration field c vanishes. We work with spherical coordinates and thus must transform the Laplace operator to spherical coordinates.

$$\vec{\nabla}^2 c = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial c}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 c}{\partial \phi^2}$$

However, this problem is radially symmetric, which means that only the partial derivative for r needs to be considered:

$$\vec{\nabla}^2 c = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c}{\partial r} \right)$$

This brings our differential equation to:

$$\begin{aligned} 0 &= D \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c}{\partial r} \right) & | \cdot \frac{r^2}{D} \\ \frac{\partial}{\partial r} \left(r^2 \frac{\partial c}{\partial r} \right) &= 0 & | \int \\ \left(r^2 \frac{\partial c}{\partial r} \right) &= A & | \cdot \frac{1}{r^2} \\ \frac{\partial c}{\partial r} &= \frac{A}{r^2} & | \int \\ c(r) &= -\frac{A}{r} + B \end{aligned}$$

To determine the constants A and B we are given the boundary condition: $c(r \rightarrow \infty) = c_\infty$, $c(r = b) = 0$

$$\begin{aligned} 0 &= -\frac{A}{b} + B \\ -Ab &= -A \\ c(r) &= -\frac{Bb}{r} + B \\ \lim_{r \rightarrow \infty} c(r) &= B = c_\infty \\ c(r) &= -\frac{c_\infty b}{r} + c_\infty \\ j &= -D \vec{\nabla} c \\ &= -\frac{D c_\infty b}{r^2} \vec{e}_r \end{aligned}$$

From this we can determine the concentration current and finally the resulting association rate:

$$I = 4\pi r^2 |j| = 4\pi D c_\infty b$$
$$k_{on} = \frac{I}{c_\infty} = 4\pi D b$$